

Varun Pabreja

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EDUCATION

University of California, San Diego

Bachelor of Science in Data Science, Economics, Cognitive Science

GPA - 3.96 | Provost Honors | 10X Teaching Assistant

San Diego, CA

September 2022 – May 2026

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, R, MATLAB, JavaScript, TypeScript

Frameworks: React, Node.js, JUnit, REST API, FASTAPI

Developer Tools: Git, GCP, AWS, VS Code, Docker, Tableau, Power BI, Anaconda, IntelliJ, Eclipse

Libraries: Pandas, NumPy, Matplotlib, Scikit-Learn, Flask, Streamlit, PySpark, TensorFlow, PyTorch, JAX

Concepts : Artificial Intelligence, Machine Learning, Deep Learning, Data Science, Data Engineering, Data Visualization, Data Analysis, CI/CD

EXPERIENCE

Data Scientist

IDC Digital

June 2025 – September 2025

Dubai, UAE

- Engineered a GCP DocumentAI-based Order Verification System to cut line item anomaly detection time by 70%
- Developed an availability-aware NLP engineer recommendation system, reducing assignment costs by 40%
- Built an enterprise RAG system using ChromaDB, MiniLM embeddings, Gemini LLM APIs, and 100-200 token chunks, cutting down response times by 60%

Data Scientist

FCSC

June 2024 – September 2024

Dubai, UAE

- Implemented DBSCAN clustering using Python to classify & predict trajectories of 100K+ vessels at 70% accuracy
- Spearheaded scalable Python, SQL pipelines analyzing 17 TB of AIS data, reducing KPI forecasting costs by 60%
- Developed 3 new trade indicators for the UAE Government, reducing report release time by 66%
- Developed an AI Data Visualization Agent using a Python, Flask back-end and a Next.js, React front-end, enhancing prediction accuracy by 22% by integrating adaptive query optimization

PROJECTS

AgntCity Times [SanDHacks 2026 Winner] | Python, FastAPI, NATS, PostgreSQL, Docker

January 2026

- Architected a supervisor-worker multi-agent AI news agency using NATS messaging with idempotent task IDs reducing dropped tasks by 90%.
- Optimized concurrency and batching to process 200+ posts per day into 10+ daily articles, sustaining throughput of 80+ posts per minute.
- Designed a PostgreSQL JSONB persistence layer for raw posts and agent artifacts, enabling replayable runs and faster retrieval to reduce article fetch latency by 76%.
- Implemented MCP server controls to lower token usage by 45% and reduce p95 model retrieval latency to 1.8s.

ClassConnect | Supabase, PostgreSQL, Python, Flask, Rest API, React, Next

September 2025

- Engineered and deployed a full-stack social platform for college students increasing active user connections by 45%
- Developed a database management system using Supabase, PostgreSQL and dynamic webpages using React, Next
- Implemented similarity-based matching algorithms using Python to match over 10K students with 88% accuracy
- Implemented secure authentication and role-based access, supporting 8K+ concurrent users

Stellantis Engineer-Ticket System | TypeScript, Python, PostgreSQL, Docker, React, AWS EC2

September 2025

- Built a full-stack engineer optimization system using TypeScript + FastAPI to model engineer efficiency and availability, raising assignment accuracy by 45%
- Designed a Python, PostgreSQL data pipeline aggregating engineer metrics, reducing query latency by 55%
- Developed a similarity-based engine for engineer-ticket assignment, cutting manual assignment costs by 60%
- Containerized development using Docker and deployed on AWS EC2 with load balancing, achieving 99% uptime